

Implementation of Cloud Based Live Streaming for Surveillance

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Abstract—Rapid technological growth made surveillance as most promising application domain. With great extent of smart city most of the things are controlled by internet. Security is one of the applications that everyone needs to be controlled remotely. This paper presents cloud based surveillance system for live video streaming that can be surveillance from anywhere and anytime. This system provides the live streaming by using cloud; Raspberry Pi 2 module and FFMPEG based USB Camera.

Index Terms—surveillance, cloud computing, web services and live streaming.

I. INTRODUCTION

Now a day crime rate has been increased and so everyone is concerned about their security and so people are moved towards smart surveillance system as technology is briskly growing. Surveillance system can be local network based, IP based or cloud based. Local network based surveillance system is low cost system with use of local network such as Wi-Fi. It is not able to cover multiple areas at a time. IP based system connects to internet via IP address. It provides good quality but needs basic knowledge of networking. Cloud computing gives a prospect for the flexible computing. This system is useful for any user and does not require any advanced computer knowledge. This will be very useful for security as well as monitoring purpose.

The proposed work has ability to control and live video streaming over the internet via cloud computing. Streaming and controlling is possible from anywhere over the world as well as can upload the data to cloud. This paper is formed as follows Section II presents the Cloud computing and web services ,Section III presents Sysem architecture, Section IV shows the working. Section V explains the system testing, finally the paper is concluded with Section IV.

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II. CLOUD COMPUTING AND WEB SERVICES

A. Cloud Computing

Cloud is a system that provides interconnected virtual computers that are dynamically facilitates one or more computer resources. It is like link between service provider and consumer.A private cloud was setup to provide Infrastructure as a Service (IaaS) using Open Stack [1]. Open Stack is a cloud operating system that manages a network of virtual machines and offers computing resources on demand [1].IaaS provides storage and computer or networking resources so that consumer can execute operating system, application or any other software. Proposed system uses Amazon EC2 cloud.

To utilize Amazon EC2, first Amazon Machine Image (AMI) is created[2-6]. AMI simply contains application program, operating system and configuration setting. This AMI is sent to storage at cloud. Storing capacity can be increased or decreased in real time. As user uses resources billing of their consumption takes place[7].

B. Web Services

Web service integrates web based applications. Different applications can communicate to each other with web services. Web services may have multiple operating system and programming language. Proposed system uses Amazon Web Services (AWS)[8].AWS provides on demand computing resources and services in cloud. Capacity and pay-as-you-go are features of AWS.

III. SYSTEM ARCHITECTURE

A system consist of Raspberry Pi 2 module, stepper motor, ULN 2008 IC, DC Motor, USB Camera (C170) and Wi-Fi Module. Raspberry Pi2 is a single board computer consist of ARM7 processor, 40 GPIO pin, 4 USB port, HDMI port, micro SD card slot, Ethernet socket and 3.5mm audio jack. Stepper motor is a device to convert electrical signal into mechanical movement. ULN 2008 IC is used to drive stepper motor. It is a high voltage IC. DC motor is L shaped motor that provides 150 rpm output. USB camera (C170) feeds image to network in real time. It is connected by simple USB port. Captured image can be saved, stored or sent to remote location. It captures video with resolution of 1024×768 pixels. Wi-Fi adaptor is used to connect the Raspberry Pi2 module to internet. Fig. 1 shows the block diagram of proposed system.

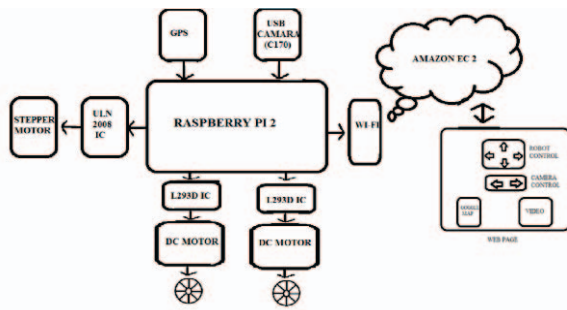


Fig. 1. Block Diagram of Proposed System

It is clear from the Fig that stepper motor is connected to Raspberry Pi module by ULN 2008 driver IC. Wheels are connected to Raspberry Pi2 via DC motor. USB Camera is attached. Web page shows video output and control buttons to navigate the module as well as movement of camera.

IV. WORKING

As the system is switched on camera starts recording the video, saves it and sends it to remote location i.e. Amazon EC2 by using Wi-Fi adaptor. Raspberry Pi2 communicates to cloud via server client model. At client side client opens the web page and request the server via entering valid IP address into address bar of web page. User can watch live video from anywhere in the world. Motion of system can be controlled via pressing the control buttons that are provided on web page. Also the movement of camera is possible with control buttons available on webpage.



Fig. 2. Data Flow from USB Camera to Web Page

Fig. 2 shows data flow from USB Camera to web page. First data is collected from USB Camera and sent to Raspberry Pi2 via USB port. Raspberry Pi2 sends this data to cloud using Wi-Fi adaptor with use of server-client model. This video can be watched lively on webpage.

V. SYSTEM TESTING AND RESULTS

Testing was carries out at different levels. Testing of connection between Raspberry Pi2 and different component like camera, Dc motor and stepper motor are successfully passed. Testing of whether system is connecting to internet or not is successfully passed.

A. Problems faced while testing

Some problems that are faced during test are :

1. Slow internet connection due to poor signal strength in some area.
2. Problem to establish connection with server and upload video to cloud.

B. Results

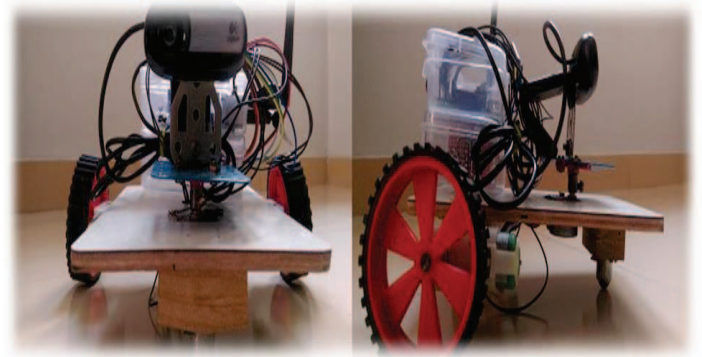


Fig. 3. Photo of hardware setup of proposed system

Fig. 3 shows the image of hardware setup of proposed module. Fig. 4 shows video frame transmitted from raspberry pi to server and Fig. 5 shows screen shot of web page where live streaming can be observed in real time from any place in the world.

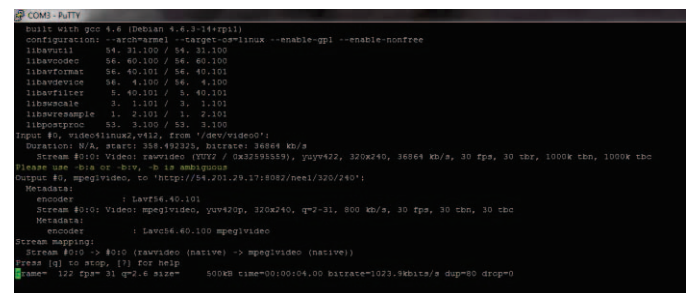


Fig. 4. Video frame transmitted to web server

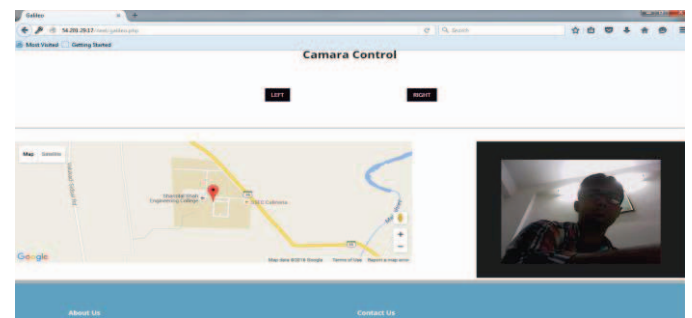


Fig. 5. screen shot of live surveillance

C. Advantages of the System

1. System is portable and can be used in many places
2. Low cost
3. Common man with basic knowledge of computer can operate it easily
4. Controlling of camera movement and navigation of whole module from web page is possible with PC as well as Android based mobile phone.

VI. CONCLUSION

Surveillance system is available with various features. Selection is based on various factors such as cost, video quality etc. Proposed system is cost effective as well as user friendly. It has application in different fields like military, defenses, house, office and environment monitoring. System can be enhanced by using face detection and recognition to follow a particular person like children below 4 years so that they are continuously in front of our eyes.

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